

# **Inflation Inequality Across the Euro Area**

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# What We Do

- ▶ Inflation inequality across households increases when relative-price shocks interact with heterogeneous household baskets.
- ▶ We construct monthly indices by income, age, and residence for all 20 euro-area countries, calibrated to official HICP weights.
- ▶ We simulate the effects of temporary VAT cuts, gas and electricity price measures, and fuel rebates on average inflation and inflation inequality.
- ▶ Our approach is close to Jaravel's distributional CPI methodology for the United States.

Source: Jaravel (2026), *Distributional Consumer Price Indices*; author's D-CPI income-quintile series.

# I Main Messages

- ▶ The euro-area inflation gap was **0.9 point**, but cross-country differences were large; age and residence also identify vulnerable households.
- ▶ Housing energy and food widened the inflation gap, while transport partly offset it; production networks added indirect exposure.
- ▶ Unconventional fiscal policies lowered cumulative average inflation by **1.9 points** and the Q1–Q5 gap by **1.1 points**.
- ▶ The distributional effect came mainly from measures on housing energy: gas and electricity measures reduced the inflation gap by **1.4 points**, while fuel measures increased it by **0.2 point**.

## Linking Official Prices to Household Baskets

- ▶ We link monthly Eurostat HICP prices and official item weights to HBS expenditure shares through a harmonized COICOP bridge.
- ▶ HBS tabulations identify baskets by income, age, and residence; microdata add household expenditures, disposable income, demographics, and survey weights.
- ▶ RAS balancing distributes official HICP weights across groups; chained Laspeyres indices preserve cross-country comparability and consistency with headline inflation.

$$P_{g,y,m} = \sum_{j \in \mathcal{J}_y} \omega_{j,g,y} \frac{p_{j,y,m}}{p_{j,y,0}}.$$
$$\underbrace{\sum_g M_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}}}_{\text{product margin}}, \quad \underbrace{\sum_{j \in \mathcal{J}_y} M_{j,g,y} = s_{g,y}}_{\text{group expenditure-share margin}}.$$

- ▶ Interpretation: baskets are held fixed between HBS waves, so the baseline abstracts from substitution and within-group heterogeneity.

## Household-Level Price Indices

- ▶ We treat each household as an elementary group and combine its 2020 expenditure shares with monthly HICP price changes.

$$P_{i,y,m} = \sum_{j \in \mathcal{J}_{i,2020,y}} \omega_{j,i,2020}^{\text{HBS}} \frac{p_{j,y,m}}{p_{j,y,0}}.$$

- ▶ Individual indices measure inflation heterogeneity within income, age, and residence categories, which broad group averages conceal.
- ▶ This within-category dispersion was substantial during the 2021–2023 inflation surge.

# Low-Income Households Faced Higher Inflation During the Energy Shock



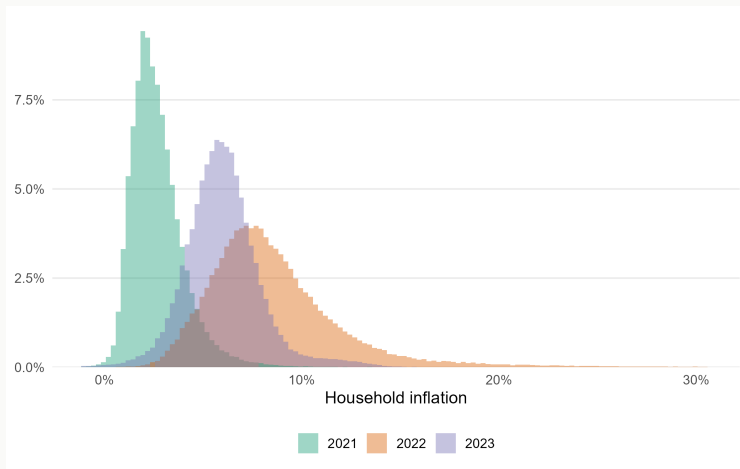
Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

# Older Households Faced Higher Inflation



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

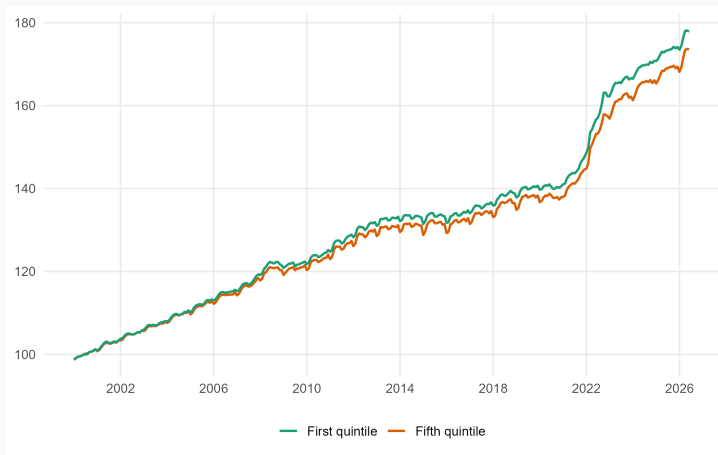
# Household Groups Hide Household-Level Dispersion



Source: Eurostat HICP and HBS microdata; authors' calculations. Units: annual household inflation (x) and household share (y), in percent.

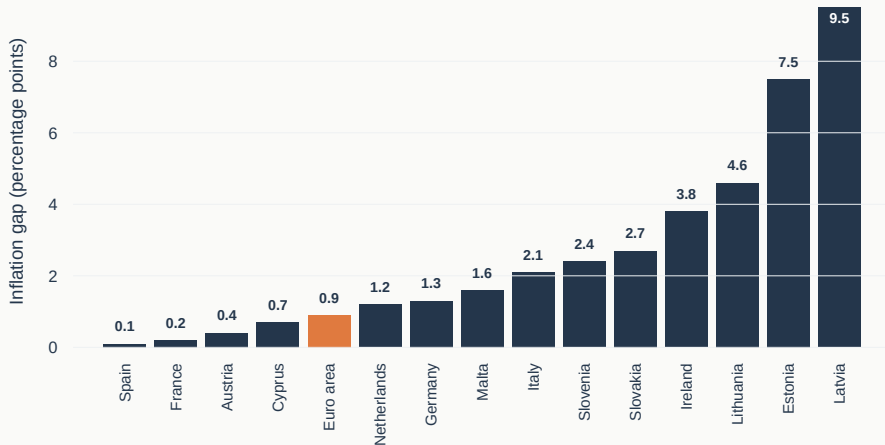


# Inflation Gaps Are Episodic Rather Than Persistent



Source: Eurostat HICP and HBS; authors' calculations. Units: price index, 2000=100.

# Heterogeneity in Inflation Gaps Across Countries



Source: Eurostat HICP and HBS; authors' calculations. Units: cumulative Q1–Q5 inflation gap, June 2021–June 2023, in percentage points. Portugal and Finland excluded.

## Energy and Food Widened the Gap; Transport Offset It

| Product group                    | Headline    | Q1          | Q5          | Gap        |
|----------------------------------|-------------|-------------|-------------|------------|
| Food and non-alcoholic beverages | 4.3         | 5.2         | 4.5         | 0.7        |
| Alcohol, tobacco and narcotics   | 0.5         | 0.7         | 0.4         | 0.2        |
| Clothing and footwear            | 0.3         | 0.3         | 0.4         | -0.1       |
| Housing, rentals and repairs     | 1.2         | 0.4         | 0.6         | -0.2       |
| Housing energy                   | 1.5         | 6.9         | 5.3         | 1.5        |
| Transport                        | 1.9         | 1.6         | 2.8         | -1.2       |
| Other                            | 5.0         | 0.2         | 0.3         | -0.1       |
| <b>Total</b>                     | <b>14.7</b> | <b>15.2</b> | <b>14.4</b> | <b>0.9</b> |

Source: Eurostat HICP and HBS; authors' calculations. Units: contributions to cumulative inflation, June 2021–June 2023, in percentage points.

# Three Instruments Lowered Retail Energy Prices

- ▶ **Price caps and tariff shields** limited household electricity and gas prices.  
*Exemple : en France, tarifs réglementés du gaz gelés à leur niveau d'**octobre 2021** dès novembre 2021 et hausse du tarif réglementé de l'électricité plafonnée à **4%** en février 2022.*
- ▶ **Temporary VAT and excise-tax reductions** lowered consumer energy prices.  
*Exemple : en France, TICFE sur l'électricité ramenée de **22,5 EUR/MWh** à **1 EUR/MWh** de février 2022 à janvier 2023.*
- ▶ **Fuel rebates and regulated discounts** reduced prices at the pump.  
*Exemple : en France, remise de **18 centimes par litre** d'avril à août 2022, portée à **30 centimes** de septembre à mi-novembre 2022.*

Source: National policy sources; Sgaravatti et al.; authors' calculations.

# Defining the Price-Policy Counterfactual

- ▶ The counterfactual removes these price-based interventions component by component while retaining observed prices for unaffected products.

## Example: reversing a temporary VAT cut

$$\tilde{p}_{j,t} = p_{j,t} \frac{1 + \tau_{j,t}^0}{1 + \tau_{j,t}^1}, \quad \tau_{j,t}^1 = \text{policy rate}, \quad \tau_{j,t}^0 = \text{no-policy rate}.$$

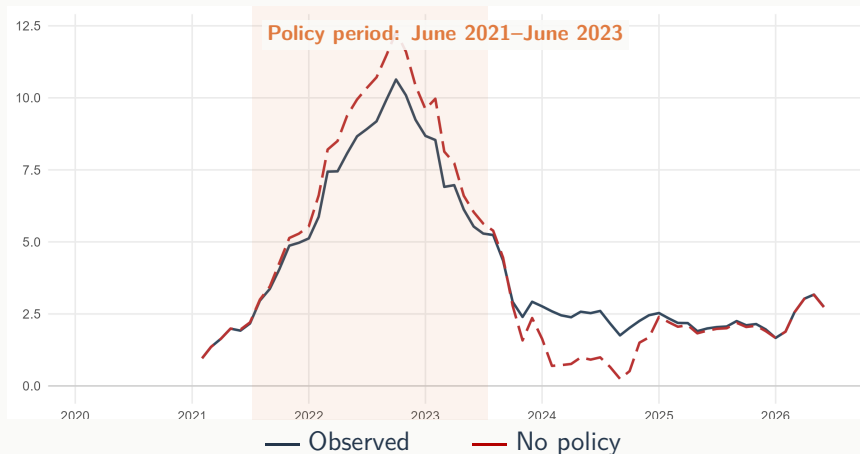
*Belgique : une TVA ramenée de 21% à 6% implique*

$$\tilde{p}_{j,t} = p_{j,t} \frac{1.21}{1.06} \simeq 1.142 p_{j,t}.$$

$j$  = électricité avec  $t$  = mars 2022–décembre 2023.

Source: National policy sources, Sgaravatti et al.; authors' calculations.

# Unconventional Fiscal Policies Lowered Inflation by 1.9 pp



Source: Eurostat and national statistical institutes; authors' calculations. Units: year-on-year inflation, in percent; January 2021–July 2026.

# Unconventional Fiscal Policies Narrowed the Inflation Gap by 1.1 pp



Source: Eurostat and national statistical institutes; authors' calculations. Units: no-policy year-on-year inflation, in percent; Jan. 2021–July 2026.

# Conclusion

- ▶ The euro-area inflation gap was **0.9 point**, but cross-country differences were large; age and residence also identify vulnerable households.
- ▶ Housing energy and food widened the inflation gap, while transport partly offset it; production networks added indirect exposure.
- ▶ Unconventional fiscal policies lowered cumulative average inflation by **1.9 points** and the Q1–Q5 gap by **1.1 points**.
- ▶ The distributional effect came mainly from measures on housing energy: gas and electricity measures reduced the inflation gap by **1.4 points**, while fuel measures increased it by **0.2 point**.



# Appendix

# Effects of Unconventional Fiscal Policies

| Country          | Fiscal cost<br>(EUR bn) | No-policy<br>inflation | Avg. effect<br>(pp) | Inflation gap<br>(pp) | Effect of fiscal policies<br>(pp) |
|------------------|-------------------------|------------------------|---------------------|-----------------------|-----------------------------------|
| Austria          | 3.8                     | 18.4                   | -1.2                | 1.1                   | -0.7                              |
| Estonia          | 0.2                     | 33.1                   | -0.1                | 7.6                   | -0.1                              |
| France           | 80.5                    | 16.9                   | -4.7                | 2.4                   | -2.2                              |
| Germany          | 60.2                    | 16.5                   | -0.8                | 2.2                   | -0.9                              |
| Ireland          | 1.4                     | 15.0                   | -0.1                | 4.0                   | -0.2                              |
| Italy            | 36.2                    | 16.4                   | -0.6                | 2.4                   | -0.3                              |
| Latvia           | 0.7                     | 28.9                   | 0.0                 | 10.2                  | -0.7                              |
| Netherlands      | 34.8                    | 19.5                   | -2.6                | 2.2                   | -0.9                              |
| Portugal         | 5.9                     | 17.0                   | -2.9                | -0.8                  | -1.5                              |
| Slovakia         | 2.3                     | 28.6                   | -3.3                | 4.8                   | -2.1                              |
| Spain            | 23.9                    | 14.3                   | -2.5                | 1.3                   | -1.2                              |
| <b>Euro area</b> | <b>268.0</b>            | <b>16.6</b>            | <b>-1.9</b>         | <b>2.0</b>            | <b>-1.1</b>                       |

- Negative effects indicate that unconventional fiscal policies reduced inflation or the Q1–Q5 gap.

Source: Eurostat and national statistical institutes; authors' calculations. Units: fiscal cost in EUR billions; cumulative inflation in percent and policy effects in percentage points, June 2021–June 2023.

## Rural Households Faced Higher Inflation at the Peak



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

## Direct Exposure Explains Most of the Energy-Shock Gap

| Country          | Total      | Direct     | Indirect   | Gap        | Gap direct | Gap indirect |
|------------------|------------|------------|------------|------------|------------|--------------|
| Austria          | 3.3        | 1.9        | 1.4        | 0.7        | 0.5        | 0.2          |
| Belgium          | 4.6        | 2.6        | 1.9        | 1.9        | 1.7        | 0.2          |
| Germany          | 4.1        | 3.0        | 1.1        | 1.5        | 1.4        | 0.1          |
| Estonia          | 4.4        | 2.1        | 2.3        | 2.2        | 0.8        | 1.4          |
| Spain            | 3.1        | 2.2        | 0.9        | 0.2        | 0.1        | 0.1          |
| France           | 3.2        | 2.3        | 0.8        | 0.6        | 0.6        | 0.0          |
| Italy            | 5.1        | 3.4        | 1.7        | -0.5       | -0.4       | -0.1         |
| Latvia           | 4.3        | 3.1        | 1.2        | 3.3        | 2.3        | 1.0          |
| Netherlands      | 3.7        | 2.9        | 0.8        | 1.4        | 1.4        | 0.0          |
| Portugal         | 3.0        | 2.0        | 1.1        | 1.1        | 0.9        | 0.2          |
| Slovakia         | 2.4        | 1.7        | 0.7        | 0.5        | 0.5        | 0.0          |
| <b>Euro area</b> | <b>3.9</b> | <b>2.7</b> | <b>1.1</b> | <b>0.8</b> | <b>0.7</b> | <b>0.1</b>   |

- ▶ A uniform **40%** energy-price shock raises the euro-area Q1–Q5 gap by **0.8 point**; direct exposure accounts for **0.7 point**.

Source: Eurostat HICP, HBS and FIGARO; authors' calculations. Units: simulated cumulative inflation effects, June 2021–June 2023, in percentage points.

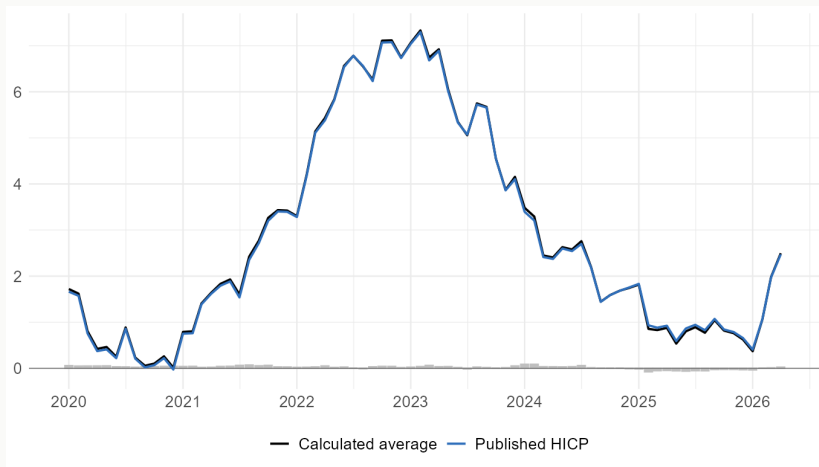
## Energy Price Measures Account for the Gap Reduction

| Product group                     | No-policy   | Q1          | Q5          | Gap        | Policy effect |
|-----------------------------------|-------------|-------------|-------------|------------|---------------|
| Food and non-alcoholic beverages  | 4.6         | 5.1         | 4.4         | 0.7        | 0.0           |
| Alcohol, tobacco and narcotics    | 0.5         | 0.6         | 0.4         | 0.2        | 0.0           |
| Clothing and footwear             | 0.3         | 0.3         | 0.4         | -0.2       | 0.0           |
| Housing, rentals and repairs      | 0.5         | 0.4         | 0.6         | -0.2       | 0.0           |
| Water, electricity, gas and fuels | 7.8         | 9.7         | 6.8         | 2.9        | -1.4          |
| Transport                         | 2.5         | 1.7         | 3.1         | -1.4       | 0.2           |
| Other                             | 0.3         | 0.2         | 0.4         | -0.1       | 0.0           |
| <b>Total</b>                      | <b>16.6</b> | <b>18.0</b> | <b>16.1</b> | <b>2.0</b> | <b>-1.1</b>   |

- ▶ Lower gas and electricity prices reduced the gap by **1.4 points**; fuel measures increased it by **0.2 point**.

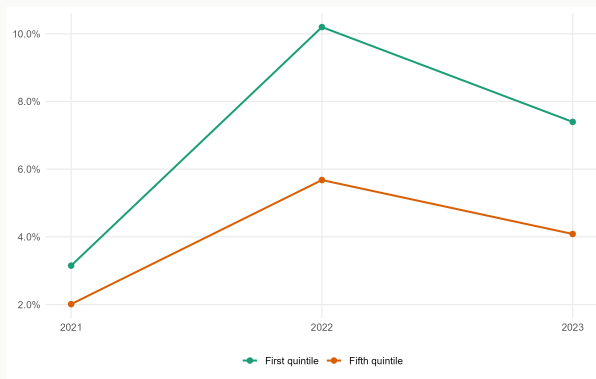
Source: Eurostat and national statistical institutes; authors' calculations. Units: product contributions to cumulative inflation and policy effects, June 2021–June 2023, in percentage points.

# RAS-Calibrated Inflation Tracks the Published HICP



Source: INSEE, Eurostat; authors' calculations. Units: year-on-year inflation, in percent.

# Inflation Burden Is Larger for Low-Income Households



- ▶ The Q1 burden reached **3.1%, 10.2%, and 7.4%** of net income in 2021–2023, versus **2.0%, 5.7%, and 4.1%** for Q5.

Source: Eurostat HICP and HBS microdata; authors' calculations. Units: annual inflation burden, in percent of household net income.

# The Paper Isolates the Household Expenditure Channel

- ▶ **Expenditure:** households buy different baskets, so relative-price shocks generate different inflation rates. This is the channel measured here.
- ▶ **Income:** wages, pensions, and transfers adjust at different speeds, producing unequal changes in real disposable income.
- ▶ **Wealth:** inflation erodes nominal assets and liabilities, transferring wealth from creditors to debtors.
- ▶ Our indices capture heterogeneity in expenditure shares within common COICOP price changes.
- ▶ They do not capture household-specific prices paid, product quality, within-category substitution, or the income and wealth channels.

Source: Main text; presentation framework adapted from the seminar PowerPoint.



# Inflation Gaps Differ Sharply Across Countries

| Country          | Total inflation | Q1 inflation | Q5 inflation | Q1–Q5 gap  |
|------------------|-----------------|--------------|--------------|------------|
| Austria          | 17.2            | 17.5         | 17.1         | 0.4        |
| Belgium          | 12.3            | 11.7         | 12.8         | -1.1       |
| Croatia          | 21.4            | 21.1         | 21.7         | -0.6       |
| Cyprus           | 12.0            | 12.7         | 12.0         | 0.7        |
| Estonia          | 33.0            | 38.7         | 31.2         | 7.5        |
| Finland          | 12.5            | 11.5         | 12.7         | -1.2       |
| France           | 12.2            | 12.3         | 12.1         | 0.2        |
| Germany          | 15.6            | 16.5         | 15.2         | 1.3        |
| Greece           | 14.7            | 15.1         | 14.5         | 0.6        |
| Ireland          | 14.9            | 17.2         | 13.3         | 3.8        |
| Italy            | 15.8            | 17.2         | 15.1         | 2.1        |
| Latvia           | 28.9            | 36.0         | 26.5         | 9.5        |
| Lithuania        | 30.4            | 33.6         | 29.0         | 4.6        |
| Luxembourg       | 11.4            | 11.0         | 11.1         | -0.2       |
| Malta            | 12.7            | 13.8         | 12.2         | 1.6        |
| Netherlands      | 16.9            | 17.7         | 16.5         | 1.2        |
| Portugal         | 14.2            | 13.1         | 15.4         | -2.3       |
| Slovakia         | 25.4            | 26.7         | 24.0         | 2.7        |
| Slovenia         | 18.1            | 19.9         | 17.5         | 2.4        |
| Spain            | 11.8            | 11.8         | 11.7         | 0.1        |
| <b>Euro area</b> | <b>14.7</b>     | <b>15.2</b>  | <b>14.4</b>  | <b>0.9</b> |

Source: Eurostat HICP and HBS; authors' calculations. Units: cumulative inflation and Q1–Q5 gap, June 2021–June 2023, in percentage points.

# Administered Prices Drive Most of the Gap Reduction

| Policy category                  | Fiscal cost<br>(EUR bn) | Average inflation<br>effect (pp) | Q1–Q5 gap<br>effect (pp) |
|----------------------------------|-------------------------|----------------------------------|--------------------------|
| <i>Instrument classification</i> |                         |                                  |                          |
| VAT cuts                         | 44.5                    | -0.5                             | -0.2                     |
| Administered prices              | 161.0                   | -1.1                             | -0.9                     |
| Fuel subsidies and rebates       | 25.4                    | -0.2                             | 0.1                      |
| Other calibrated layer           | 37.1                    | -0.2                             | -0.2                     |
| <b>Total</b>                     | <b>268.0</b>            | <b>-1.9</b>                      | <b>-1.1</b>              |
| <i>Targeted vs. untargeted</i>   |                         |                                  |                          |
| Untargeted price measures        | 235.2                   | -1.5                             | -0.8                     |
| Targeted price measures          | 32.8                    | -0.4                             | -0.4                     |
| <b>Total</b>                     | <b>268.0</b>            | <b>-1.9</b>                      | <b>-1.1</b>              |

- Negative values indicate that policies reduced inflation or the inflation gap.

Source: Bruegel; national sources for VAT food products; authors' calculations. Units: fiscal cost in EUR billions; average-inflation and Q1–Q5-gap effects, June 2021–June 2023, in percentage points.